

CPV 2lss Weaver Project Status

Local project summary

August 21, 2026

Executive Summary

The current project baseline is now `eventmlp_fullsplit`, batch 512, learning rate `1e-4`. The model choice is close rather than decisive on held-out AUC: `largerfc_fullsplit` has 0.64848, while `eventmlp_fullsplit` has 0.64762. The reason to prefer `eventmlp_fullsplit` is the smaller overtraining signature. Its train/test AUC gap is about 0.03256, compared with about 0.03984 for `largerfc_fullsplit`.

The recommended current configuration is:

- Data config: `data/cpv_part_2lss_fullweight_bdtvars.yaml`
- Network config: `data/cpv_part_2lss_eventmlp_model.py`
- Model variant: `MODEL_VARIANT=eventmlp`
- Model tag: `MODEL_TAG=eventmlp_fullsplit_lr1em4`
- Loss mode: class-balanced event-weighted cross entropy
- Loss weight branch: `abs(weight)`
- Optimizer: `ranger`
- Batch size: 512
- Learning rate: `1e-4`
- Epochs: 50
- Split: train `event_mod5 < 3`, validation `event_mod5 == 3`, held-out/test `event_mod5 == 4`

This should be treated as a good starting point, not as a frozen optimum, when adding 2024 data. The current era feature block only encodes 2022, 2022EE, 2023, and 2023BPix. Adding 2024 means updating the era flags and regenerating preprocessing, then rerunning at least a small hyperparameter sanity scan.

Default Files and Launchers

Role	Current file or default
Training launcher	<code>execution_scripts/launcher_cpv_part_2lss.sh</code>

Inference/test launcher	<code>execution_scripts/launcher_cpv_part_2lss_test.sh</code>
Slurm wrapper	<code>execution_scripts/submit_one_job.sh</code>
Default data snapshot	<code>/work/abelvede/multilepton-analysis/neuralnetwork/snapshots/cpv_part_2lss/2026-05-28</code>
Default data config	<code>data/cpv_part_2lss_fullweight_bdtvars.yaml</code> via <code>WEIGHT_MODE=fullweight_bdtvars</code>
Alternative data config	<code>data/cpv_part_2lss_cpmodelweight_bdtvars.yaml</code> via <code>WEIGHT_MODE=cpmodelweight_bdtvars</code>
Default model config	<code>data/cpv_part_2lss_eventmlp_model.py</code> via <code>MODEL_VARIANT=eventmlp</code>
Default model tag	<code>eventmlp_fullsplit_lr1em4</code> ; full training name ends with <code>fetch1_batch512_lr0p0001</code>
Larger-FC model config	<code>data/cpv_part_2lss_model.py</code> via <code>MODEL_VARIANT=largerfc</code>
Wide Event-MLP config	<code>data/cpv_part_2lss_eventmlp_wide_model.py</code> via <code>MODEL_VARIANT=eventmlpwide</code>
Object-only config	<code>data/cpv_part_2lss_objectonly_model.py</code> via <code>MODEL_VARIANT=objectonly</code>
Evaluation helper	<code>evaluation/evaluate_cpv_part_2lss.py</code>
Training outputs	<code>result/cpv_part_2lss/<training_name>/fullmodel*</code>
Prediction outputs	<code>result/cpv_part_2lss/<training_name>/predict_output_<split>*.root</code>
Training logs	<code>logs_full_model/log_fullmodel_<training_name>.log</code>
Test logs	<code>logs_full_model/log_test_<training_name>_<split>.log</code>
Plots and summaries	<code>evaluation_output/<training_name>/summary.txt</code> , ROC plots, confusion matrices, loss curves

The launcher builds training names from the CPV task name, weight mode, loss tag, model tag, fetch step, batch size, and learning-rate tag. The default launcher values are:

- `BATCH=512`
- `START_LR=1e-4`
- `NUM_EPOCHS=50`
- `WEIGHT_MODE=fullweight_bdtvars`
- `LOSS_TAG=classbalancedloss`
- `MODEL_VARIANT=eventmlp`
- `MODEL_TAG=eventmlp_fullsplit_lr1em4`

Data and Split Status

The current full-weight BDT-variable data config uses:

- Labels: `is_cp_even: cp_label == 0, is_cp_odd: cp_label == 1`
- Loss weight: `loss_weight: np.abs(weight)`

- Holdout selection in the YAML: `selection: event_mod5 != 4`
- Test-time selection in the YAML: `test_time_selection: event_mod5 == 4`
- Additional training split: `-extra-selection-train 'event_mod5 < 3'`
- Additional validation split: `-extra-selection-val 'event_mod5 == 3'`
- Object inputs: 8 jets, 3 lepton/MET objects, masks, four-vectors, b-tag categories, hadronic-top flags
- Event scalar inputs: CPV angles/kinematics, `cpv_b1` through `cpv_b4`, minimum lepton-jet distances, average jet-pair `dR`, and era one-hot flags

The `cpmodelweight_bdtvars` config is structurally similar but changes the loss weight to `np.abs(cp_model_weight)`. In the measured runs, the full event weight configuration was better for the final baseline.

Model Variants

All model variants share the same main object backbone: a `ParticleTransformerTagger` over jet and lepton/MET objects. The common default backbone uses:

- `embed_dims=(128, 512, 128)`
- `pair_embed_dims=(64, 64, 64)`
- `num_heads=8`
- `num_layers=8`
- `num_cls_layers=2`
- `activation="gelu"`

largerfc

File: `data/cpv_part_2lss_model.py`

`largerfc` runs the object transformer, concatenates the normalized event-level scalar features directly to the transformer representation, and then applies a larger fully connected classification head:

```
fc_params=((256, 0.1), (128, 0.1), (64, 0.1))
```

This variant remains important because it has the highest held-out AUC found locally. It is simple and avoids adding a separate preprocessing subnetwork for the scalar event variables. It is no longer the default baseline because the corresponding train/test gap is larger than for `eventmlp_fullsplit`.

eventmlp

File: `data/cpv_part_2lss_eventmlp_model.py`

`eventmlp` also uses the same object transformer, but it first passes event-level scalar features through a small MLP:

```
event_embed_dims=(64, 64), event_dropout=0.1
```

The embedded event representation is then concatenated to the transformer output, followed by a smaller classifier head:

```
fc_params=((128, 0.1), (64, 0.1))
```

This is now the default baseline. It gives a held-out AUC very close to `largerfc`, but with less overtraining: the train/test AUC gap is about 0.03256 instead of 0.03984. This makes it the more conservative default for the next round, especially before adding 2024 data.

eventmlpwide

File: `data/cpv_part_2lss_eventmlp_wide_model.py`

This is the wider event-feature MLP variant:

```
event_embed_dims=(128, 128, 64), event_dropout=0.1
```

It did not beat the smaller event MLP or `largerfc` in the historical runs.

objectonly

File: `data/cpv_part_2lss_objectonly_model.py`

This variant ignores the event scalar inputs and classifies from the object transformer representation only. It was useful as an ablation. Its best held-out AUC was below the main event-feature models, confirming that the scalar CPV/BDT-style variables are helpful.

Run Results

Run	Held-out AUC	Train AUC	Comment
eventmlp fullsplit, batch512, lr1e-4	0.64762	0.68018	Default baseline; smaller AUC gap, about 0.03256
largerfc fullsplit, batch512, lr1e-4	0.64848	0.68832	Highest held-out AUC; larger AUC gap, about 0.03984
largerfc, batch512, lr1e-4	0.64771	0.68583	Earlier tag, similar AUC but more overtraining than <code>eventmlp</code>
largerfc fullsplit, batch1024, lr1e-4	0.64762	0.67909	Batch 1024 did not improve held-out AUC
largerfc dropout0p2, batch512, lr1e-4	0.64630	0.68982	More dropout did not help
largerfc wd1e-4, batch512, lr1e-4	0.64491	0.66167	Weight decay reduced overtraining but not best held-out AUC
eventmlp fullsplit, batch512, lr5e-5	0.64436	0.66502	Lower LR did not improve <code>eventmlp</code>

objectonly, batch512, lr1e-4	0.63838	0.65624	Event features matter
cpmodelweight eventmlp, batch512, lr0.01	0.62732	0.63501	Older high-LR run
cpmodelweight largerfc, batch512, lr0.01	0.62423	0.63259	Older high-LR run
fullweight weightedloss, batch512, lr0.01	0.61578	–	Early weighted-loss baseline

The important trend is that the serious later comparison moved away from the old `lr=0.01` trials and settled on `fullweight_bdtvars + classbalancedloss`, with `lr=1e-4`. The `largerfc` and `eventmlp` results are close. With overtraining included in the decision, the launcher default now points to `eventmlp_fullsplit`.

Current Recommendation

For the next training round without changing the dataset composition, use:

```
./execution_scripts/submit_one_job.sh ./execution_scripts/launcher_cpv_part_2lss.sh
```

or explicitly:

```
./execution_scripts/submit_one_job.sh WEIGHT_MODE=fullweight_bdtvars
MODEL_VARIANT=eventmlp MODEL_TAG=eventmlp_fullsplit_lr1e4 START_LR=1e-4 BATCH=512
./execution_scripts/launcher_cpv_part_2lss.sh
```

For producing test/train/validation prediction outputs, use:

```
./execution_scripts/submit_one_job.sh EVAL_SPLIT=test
./execution_scripts/launcher_cpv_part_2lss_test.sh
```

and repeat with `EVAL_SPLIT=train` or `EVAL_SPLIT=val` when needed. The test launcher can also take `MODEL_PREFIX` and `PREDICT_OUTPUT` if evaluating a specific epoch rather than the default model prefix.

Impact of Adding 2024

Adding 2024 data is likely to require more than changing `DATADIR`. The current configs define era one-hot flags only for:

```
era_is_2022, era_is_2022EE, era_is_2023, era_is_2023BPix
```

Before retraining with 2024, update the data configs to include the correct 2024 era categories according to the upstream `eraFlag` convention. Then regenerate the auto preprocessing configs, because the current auto-normalization constants were fitted on the previous 2022–2023 snapshot.

Suggested retuning path after adding 2024:

1. Update the YAML era flags and event feature list for 2024.

2. Regenerate or refresh auto preprocessing for `fullweight_bdtvars`; optionally do the same for `cpmodelweight_bdtvars`.
3. Train `eventmlp`, batch 512, `lr=1e-4`, as the reference.
4. Train `largerfc`, batch 512, `lr=1e-4`, as the higher-AUC but more-overtrained cross-check.
5. If compute allows, scan `lr=5e-5` and `batch=1024` for the better of those two models.
6. Re-evaluate train, validation, and held-out/test splits with the same weighted ROC and confusion matrix tooling.

Evidence Files

Key files used for this status summary. The long result names in the table above are shortened labels for these output directories.

- `evaluation_output/cpv_part_2lss_fullweight_bdtvars_classbalancedloss_eventmlp_fullsplit_lr1e-4_fetch1_batch512_lr0p0001/summary.txt`
- `evaluation_output/cpv_part_2lss_fullweight_bdtvars_classbalancedloss_largerfc_fullsplit_fetch1_batch512_lr0p0001/summary.txt`
- `evaluation_output/cpv_part_2lss_fullweight_bdtvars_classbalancedloss_largerfc_fullsplit_batch1024_fetch1_batch1024_lr0p0001/summary.txt`
- `execution_scripts/launcher_cpv_part_2lss.sh`
- `execution_scripts/launcher_cpv_part_2lss_test.sh`
- `data/cpv_part_2lss_fullweight_bdtvars.yaml`
- `data/cpv_part_2lss_model.py`
- `data/cpv_part_2lss_eventmlp_model.py`